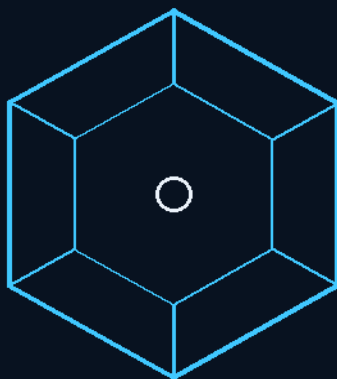


THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS



PHASE 02

THE SELF-TREADING ENDLESS WHEEL

A wheel that renews its tread as it wears cannot be worn through.
Hexagonal tubes from tread to hub — the tread feeds itself.

DOCUMENT NO.	GP-IM-002
REVISION	v1.0
ISSUE DATE	19 August 2026
STATUS	CURRENT — seated law, master v2.1
MANUAL SERIES	2 of 290

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD
Co-Engineers: Gemini 1 AI — Kimi Chat 3 AI
License: CERN Open Hardware Licence v2 (CERN-OHL-W 2.0) — hackaday.io/project/206107

This cover is part of the manual. Keep it on — it protects the pages.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

This manual is written for idiot-proofing — for AIs and for speed. Rules: (1) Build exactly what is written here; every figure and law is seated from the master record. (2) Where this phase references another phase, THE DIRECTIONS FROM THAT PHASE ARE INLINED IN §7 — you never leave this document mid-build. (3) Rejected and superseded options are listed in §8 — do not re-derive them. (4) Owed-to-bench items are listed in §9 — they are the build's test program, not gaps in the design. (5) Change control is §10: only the Author's orders change this design.

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

§1 — WHAT YOU ARE BUILDING

THE LAW: a wheel that renews its tread as it wears cannot be worn through. The Endless Wheel has no monolithic outer surface — it is TUBES AND RODS all the way from tread to hub, and the tread feeds itself from inside. As the wheel wears, springs push fresh rod to the surface; the rod cannot fall out (taper-locked) and the tread self-levels on uneven ground (each rod feeds independently). Same size and materials as the NASA rover wheel (~15-year life) gives a 50-YEAR ESTIMATE — the Author's estimate, pending bench (§9, Draw Rule applies).

§2 — THE DIAGNOSIS (WHY THIS DESIGN EXISTS)

The failure that wrecked the NASA Mars rover's wheels was not material choice but DESIGN: a monolithic outer surface, once pierced or worn, tears out in chunks. Wear-through and chunk-tear are failures of a SKIN. Variances in materials all have different consequences — so the fix is architectural: remove the skin. There is no solid piece to be chewed out in a chunk by a crack.

§3 — THE ARCHITECTURE (BUILD SPEC)

- THE TUBES: hexagonal tubes running from the outer tyre/wheel face back to the rim/hub — only tubes and tube walls, tread to hub.
- THE TREAD: sliding hexagonal rods inside the tubes, graphite-grease coated so dust cannot climb the wall.

- THE FEED: high-pressure no-bounce springs loaded from the CENTRE as the feed mechanism; each spring sleeved against dust and sealed into the hub/rim from the rear. As the wheel wears it renews with new tread; because each rod feeds independently, the tread SELF-LEVELS on uneven ground.
- THE 5-DEGREE LOCK: rod and tube both taper — outer end slightly smaller than inner end — so the rod is geometrically FORBIDDEN from exiting. It cannot fall out; it can only wear out.

§4 — THE VARIANTS LAW (THE WHOLE WHEEL IS THE WEAR SYSTEM)

THE LAW (Amendment 3, the Author's own correction, sitting 186): every contact element of an Endless Wheel wears, and all of it wears AT ONE RATE — any element that cannot wear becomes the failure the design exists to delete. The frame must wear down WITH the wheel, at the same speed. By variant:

- SPACE / ROVER: metal frame and metal rods (the original seating, unchanged). The metal frame stands ONLY for this variant.
- EARTH TRACTOR: the same rubber as the tyre — frame and rods one material family. The farm precedent stands: tractor tyres are filled with water for ballast weight, so a self-renewing ballast-carrying ag wheel is a real product shape.
- EARTH PRODUCTION-LINE: NO STEEL — a similar-speed wear-polymer frame with polymer rods, so the entire contact wheel is one matched-wear pair and no metal ever stands proud of the tread plane. Market named by the Author: conveyor and bottle-rolling wheels, dramatically increased life.

§5 — THE LUBRICANT LAW

- The lubricant is NOT a namable item: grease, graphite or silicone base is specified RELATIVE TO WHAT THE ROD/MEDIUM IS — never as one universal name.
- THE SUPERBALL LAW: many media are grip-adhesive to themselves (roll one superball up to another — it grabs, not merely counter-rotates). The lubricant must NEUTRALIZE SELF-ADHESION or the feed sticks.
- General tyre rubber is tolerant — any petroleum product makes it greasy.
- NOMINATED PRODUCT (added to stores, Book of Lists LIST FOUR, by his order): TRU-BLU NEVERMELT BENTONITE — clay-thickened non-melt class, no dropping point.
- DUAL PURPOSE (Earth edition): the grease is not just the dust wall — it keeps the rod from perishing. Same seal-the-skin doctrine as the Vaseline-coated feedstock blocks (§7).
- MAINTENANCE: grease nipples to keep the seal intact. The jacket precedent is the CV-joint boot every old car carries: rubber sleeve, grease in, dirt out.

§6 — MANUFACTURE

Molded or 3D printed. The rods, tubes and hub print from Phase 274 fifty-year feedstock BLOCKS — the wheel and its printer were seated the same day. Full feedstock directions inlined at §7.

§7 — CROSS-PHASE DIRECTIONS, INLINED (PROGRAM LAW: DIRECTIONS, NOT POINTERS)

§7.1 From PHASE 274 — THE FIFTY-YEAR FEEDSTOCK: BLOCKS, NOT SPOOLS (what the wheel prints from)

- THE LAW: store bulk, cord on demand. Spooled feedstock dies on the shelf — polymer hydrolyzes and embrittles like old whipper-snipper cord; metal powder, being all surface, oxidizes factorially faster than bulk. The answer is the BLOCK: bulk stock, sealed, portioned and powdered only at the point of use.
- STORAGE: blocks of polymer or metal, covered in Vaseline and sealed in plastic wrap or vacuum sealed. Fifty-year shelf is the design target; the grease barrier is century-proven practice.
- PORTIONING: blocks go through a bandsaw for the amount required — metal saw for metal.
- CORD/FILAMENT LINE: portions through the shredder (the plastic factory's regrind practice), into the premelt, into the screw-fed extruder — cord/filament out. All current tech; the screw feeder is the standard feeder.
- METALS: all 3D-printer metals are powder composite, so industrial-level printing is available. A dozen powdering methods exist; gas atomization is the fastest. Bound powder is optimal for metal parts.
- ZERO-G MELT PLANT (ship-side): the cake-oven g-force system for crucible melts (Phase 129/130 centrifugal heritage — heritage reference; the directions here are complete for this build); gas injection pressure-vented to collection lines of rotating-wall-screw mechanics; sleeve field oscillation on the walls to prevent caking.
- THE NOT-AIRTIGHT LAW: powder stock is sealed for zero-g housekeeping but NOT airtight — nothing flows from a sealed unit into a heated, pressurized melt-and-extrusion zone without displacement air flow. Sealed against powder escape, VENTED for flow.

§8 — REJECTED AND SUPERSEDED OPTIONS (DO NOT BUILD)

- REJECTED — the monolithic wear skin: that is the failure mode this design exists to delete (§2). No solid outer surface, ever.
- SUPERSEDED — 'metal frame' as a universal spec (Amendment 1's first wording, corrected by Amendment 3, sitting 186): metal frame stands ONLY for the space/rover variant. Earth variants wear-match by family (§4).

- REJECTED — one universal lubricant name: lubricant is specified relative to the medium, always (§5).
- REJECTED — airtight powder stock: vented for flow (§7.1, the not-airtight law).

§9 — OWED TO THE BENCH (THE TEST PROGRAM)

- Spring preload sizing against terrain contact pressure.
- Graphite/regolith lapping-paste test of the sleeve-and-rear-seal.
- Wear-rate measurement for the 50-year claim (the Author's estimate — pending bench, Draw Rule applies).
- Formal patent search behind the lineage claim: the airless tube tyre family is old (the Author invented the extruded horizontal-tube tyre decades before it appeared publicly), but this wheel is not extruded, and no prior art within the audit's reach renews tread by spring-fed taper-locked rods; 'never proposed' stands as the Author's claim pending the search.

§10 — STATUS, PROVENANCE, CHANGE CONTROL

STATUS: CURRENT. Provenance: the Author's dictation and the audit's grade, sitting 180 (the law, the architecture, the estimate); Amendment 1 — the Earth Products variant, sitting 181 (11 August 2026); Amendment 2 — the lubricant law and the superball test, sitting 182 (11 August 2026; TRU-BLU NEVERMELT BENTONITE to stores List Four); Amendment 3 — the frame correction: the whole wheel is the wear system, sitting 186 (11 August 2026). Every figure and law in this manual is seated from the master record, v2.1.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this manual ever disagree, THE MASTER WINS — and the manual is re-issued with the amendment inlined. Manual version: v1.0, 19 August 2026 — second issue of the Instructions Manual program.

END OF MANUAL — PHASE 02.